

### Overview

My research develops simple, principled methods that improve the tradeoff between foundation model quality and compute and memory efficiency. I study algorithms, hardware, and applications in lockstep.

### Education

- 2019-2025      **PhD in Computer Science**, *Stanford University*  
Research advisor: Christopher Ré  
Thesis: [Pareto-efficient sequence modeling](#)
- 2015-2019      **Jerome Fisher Management & Technology Program**, *University of Pennsylvania*  
Summa cum laude BSE in Computer Science, School of Engineering  
Summa cum laude BS in Economics (Finance Concentration), The Wharton School  
Minor in Mathematics  
Research advisors: Boon Thau Loo, Vijay Kumar, Vincent Liu

### Positions

- Starting 2027      **Assistant Professor**, *California Institute of Technology*  
Department of Computing + Mathematical Sciences
- 2025-present      **Frontier Performance Team Lead**, *Together AI*  
Working on efficient AI inference
- 2024-present      **Advisor**, *Cartesia AI, Engram, Pearl, and Fearn AI*
- 2021-2022      **Research Scientist**, *Facebook AI Research*  
Worked on private neural information retrieval
- Summer 2018      **Technology Investment Banking**, *Morgan Stanley, Menlo Park*  
Worked on the sale of Acxiom AMS to IPG for \$2.3Bn, sale of Cylance to BlackBerry for \$1.7Bn, and Sonos IPO on Nasdaq
- Summer 2017      **Software Engineering**, *Google*  
JavaScript Open Source Compiler, [GitHub closure-compiler](#)

### Selected Research Impact and Adoption

Bootleg ([Apple](#), [ZDNet](#)); Ask Me Anything ([Forbes](#), [Snorkel AI](#), [SambaNova](#), [Numbers Station](#), [Aleksa Gordić](#)); EVAPORATE ([LlamaIndex](#)); Zoology, Based, JRT, and LoLCATS ([Hugging Face](#), [NVIDIA](#), [Together AI](#), [Mamba-v2](#), [x-LSTM](#), [Gated Linear Attention](#), [AI21 Labs Jamba](#), [DeltaNet](#), [Moonshot AI Kimi Linear](#)), Monarch Mixer ([MongoDB](#), [Together AI](#), [Nomic AI](#)); ConcurrentQA ([Meta](#)); Perfect secrecy ([VentureBeat](#)); Data Wrangling ([Numbers Station](#)); KernelBench ([NVIDIA](#); [METR](#); [Sakana AI](#); [Cognition](#)); ThunderKittens ([Together AI](#), [Cruise](#), [Cursor](#), [Tencent AI](#), [Intel Corporation](#), [NVIDIA](#), [Cloudflare](#)); HipKittens ([AMD AITER #1](#), [AMD AITER #2](#), [AMD Transformer Engine #1](#)); ThunderAgent ([NVIDIA](#), [SkyRL](#))

## Awards

|      |                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>ICML DL4C Oral Award (Top 5 of 80+ accepted papers)</b><br>Hardware-Aware GPU Kernel Optimization with Minimal Supervision                        |
| 2026 | <b>ICML Spotlight Award (Top 2.2% of 24K submissions)</b><br>ThunderAgent: A simple, fast and program-aware agentic inference system                 |
| 2026 | <b>MLSys Distinguished Artifact Award</b><br>ParallelKittens: Systematic and Practical Simplification of Multi-GPU AI Kernels                        |
| 2025 | <b>ICML ES-FoMo Oral Award (Top 5 of 146 submissions)</b><br>Cartridges: Lightweight long context representations via self-study                     |
| 2025 | <b>ICLR DL4C Best Paper Award (Top 1 of 63 accepted papers)</b><br>KernelBench: Can LLMs Write Efficient GPU Kernels?                                |
| 2025 | <b>ICLR Spotlight Award (Top 5.1% of 11.7K submissions)</b><br>ThunderKittens: Simple, Fast, and Adorable AI Kernels                                 |
| 2025 | <b>Stanford Computer Science Graduate Fellowship, 1-year</b>                                                                                         |
| 2024 | <b>ICML ES-FoMo Best Paper Award (Top 1 of 83 accepted papers)</b><br>Simple linear attention language models balance the recall-throughput tradeoff |
| 2024 | <b>ICML Spotlight Award (Top 3.5% of 10K submissions)</b><br>Simple linear attention language models balance the recall-throughput tradeoff          |
| 2023 | <b>NeurIPS Outstanding Paper Award (Top 4 of 12.3K submissions)</b><br>DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models    |
| 2023 | <b>NeurIPS Oral Award (Top 1% of 1K submissions)</b><br>DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models                   |
| 2023 | <b>NeurIPS Oral Award (Top 0.5% of 12.3K submissions)</b><br>Monarch Mixer: A Simple Sub-Quadratic GEMM-Based Architecture                           |
| 2023 | <b>ICLR Spotlight Award (Top 5% of 5K submissions)</b><br>Ask Me Anything: A simple strategy for prompting language models                           |
| 2019 | <b>Stanford Graduate Fellowship, 3 years</b>                                                                                                         |
| 2019 | <b>Rhodes Scholarship National Finalist</b>                                                                                                          |
| 2019 | <b>Marshall Scholarship National Finalist</b>                                                                                                        |
| 2019 | <b>Penn Computer Science Academic Award</b><br>One graduating CS major per year                                                                      |
| 2019 | <b>Michele Huber and Bryan D. Giles Memorial Award</b><br>One graduating Jerome Fisher student per year                                              |
| 2019 | <b>Penn Computer Science Senior Engineering Capstone Project 2nd Place</b>                                                                           |
| 2019 | <b>Wharton School Summa Cum Laude (Highest Honors)</b>                                                                                               |
| 2019 | <b>Penn Engineering Summa Cum Laude (Highest Honors)</b>                                                                                             |
| 2017 | <b>Best Paper Runner Up: IEEE MARSS Conference (Top 3 papers)</b><br>Control of multiple microrobots with multiscale magnetic field superposition    |
| 2017 | <b>University of Pennsylvania Tau Beta Pi and Eta Kappa Nu</b>                                                                                       |
| 2015 | <b>International University Physics Competition (Top 20%, <a href="#">Link</a>)</b>                                                                  |

## Publications

- [1] Alexander Waitz, William Hu, Benjamin Spector, Atri Rudra, Christopher Ré, Simran Arora  
*Cylon: Asynchronous Linear Attention*  
Conference on Language Modeling (COLM) 2026
- [2] Hao Kang, Ziyang Li, Weili Xu, Xinyu Yang, Yinfang Chen, Junxiong Wang, Beidi Chen, Tushar Krishna, Chenfeng Xu, Simran Arora  
*ThunderAgent: A Simple, Fast and Program-Aware Agentic Inference System*  
International Conference on Machine Learning (ICML) 2026  
**Spotlight Award (Top 2.2% of 24K submissions)**  
[Paper Link](#) / [GitHub Link](#)
- [3] William Hu, Drew Wadsworth, Sean Siddens, Stanley Winata, Daniel Y. Fu, Ryann Swann, Muhammad Osama, Christopher Ré, Simran Arora  
*HipKittens: Fast and Furious AMD Kernels*  
Conference on Machine Learning Systems (MLSys) 2026  
[Paper Link](#) / [GitHub Link](#)
- [4] Stuart H. Sul, Simran Arora, Benjamin F. Spector, Christopher Ré  
*ParallelKittens: Systematic and Practical Simplification of Multi-GPU AI Kernels*  
Conference on Machine Learning Systems (MLSys) 2026  
**Distinguished Artifact Award**  
[Paper Link](#) / [GitHub Link](#)
- [5] Sabri Eyuboglu, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Will Tennien, Atri Rudra, James Zou, Azalia Mirhoseini, Christopher Ré  
*Cartridges: Lightweight and general-purpose long context representations via self-study*  
International Conference on Learning Representations (ICLR) 2025  
ICML ES-FoMo 2025 **Oral Award (Top 5 of 146 submissions)**  
[Paper Link](#) / [GitHub Link](#)
- [6] Anne Ouyang, Simon Guo, Simran Arora, Alex L. Zhang, William Hu, Christopher Ré, Azalia Mirhoseini  
*KernelBench: Can LLMs Write Efficient GPU Kernels?*  
International Conference on Machine Learning (ICML) 2025  
ICLR DL4C 2025 **Best Paper Award**  
[Paper Link](#) / [GitHub Link](#)
- [7] Benjamin Spector, Simran Arora, Aaryan Singhal, Daniel Fu, Christopher Ré  
*ThunderKittens: Simple, Fast, and Adorable AI Kernels*  
International Conference on Learning Representations (ICLR) 2025  
**Spotlight Award (Top 5.1% of 11.6K submissions)**  
[Paper Link](#) / [GitHub Link](#)
- [8] Michael Zhang, Simran Arora, Rahul Chalamala, Benjamin Spector, Alan Wu, Krithik Ramesh, Aaryan Singhal, Christopher Ré  
*LoLCATS: Low-rank Linearization of Large Language Models*  
International Conference on Learning Representations (ICLR) 2025  
[Paper Link](#) / [GitHub Link](#)
- [9] Jerry Liu, Jessica Grogan, Owen Dugan, Simran Arora, Atri Rudra, and Christopher Ré  
*Towards Learning High-Precision Least Squares Algorithms with Sequence Models*  
International Conference on Learning Representations (ICLR) 2025  
[Paper Link](#)

- [10] Megha Srivastava, Simran Arora, Dan Boneh  
*Optimistic Verifiable Training by Controlling Hardware Nondeterminism*,  
 Advances in Neural Information Processing Systems (NeurIPS) 2024  
 ICML ES-FoMo 2024  
[Paper Link](#) / [GitHub Link](#)
  
- [11] Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, James Zou, Atri Rudra, Christopher Ré  
*Simple linear attention language models balance the recall-throughput tradeoff*  
 International Conference on Machine Learning (ICML) 2024  
**Spotlight Award (Top 3.5% of 10K submissions)**  
 ICML ES-FoMo 2024 **Best Paper Award**  
[Paper Link](#) / [GitHub Link](#)
  
- [12] Jon Saad-Falcon, Daniel Y. Fu, Simran Arora, Neel Guha, Christopher Ré  
*Benchmarking and Building Long-Context Retrieval Models with LoCo and M2-BERT*  
 International Conference on Machine Learning (ICML) 2024  
 ICML ES-FoMo 2024  
[Paper Link](#) / [GitHub Link](#)
  
- [13] Furui Cheng, Vilém Zouhar, Simran Arora, Mrinmaya Sachan, Hendrik Strobelt, Mennatallah El-Assady  
*RELIC: Investigating Large Language Model Responses using Self-Consistency*  
 Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI) 2024  
[Paper Link](#)
  
- [14] Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, Christopher Ré  
*Zoology: Measuring and Improving Recall in Efficient Language Models*  
 International Conference on Learning Representations (ICLR) 2024  
[Paper Link](#) / [GitHub Link](#)
  
- [15] Daniel Y. Fu, Simran Arora, Jessica Grogan, Isys Johnson, Sabri Eyuboglu, Armin W. Thomas, Benjamin F. Spector, Michael Poli, Atri Rudra, Christopher Ré  
*Monarch Mixer: A Simple Sub-Quadratic GEMM-Based Architecture*  
 Advances in Neural Information Processing Systems (NeurIPS) 2023  
**Oral Award (Top 0.5% of 12.3K submissions)**  
 Daniel Y. Fu\* and Simran Arora\*, Revisiting BERT, Without Attention or MLPs, [Link](#)  
[Paper Link](#) / [GitHub Link](#)
  
- [16] Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li  
*DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models*  
 Advances in Neural Information Processing Systems (NeurIPS) 2023  
**Oral Award (Top 1% of 1K submissions)**  
**Outstanding Paper Award (Top 4 accepted papers)**  
[Paper Link](#) / [Benchmark Link](#)
  
- [17] Simran Arora, Brandon Yang, Sabri Eyuboglu, Avanika Narayan, Andrew Hojel, Immanuel Trummer, Christopher Ré  
*Language Models Enable Simple Systems for Generating Structured Views of Data Lakes*  
 Proceedings of the VLDB Endowment (PVLDB) 2023.  
[Paper Link](#) / [GitHub Link](#)

- [18] Simran Arora, Patrick Lewis, Angela Fan, Jacob Kahn, Christopher Ré  
*Reasoning over Public and Private Data in Retrieval-Based Systems*  
 Transactions of the Association for Computational Linguistics (TACL) 2023  
 AAAI KnowledgeNLP 2023 **Oral Award (Top 15% of accepted papers)**  
[Paper Link](#) / [GitHub Link](#)
- [19] Simran Arora, Avanika Narayan, Mayee F. Chen, Laurel Orr, Neel Guha, Kush Bhatia, Ines Chami, Frederic Sala, Christopher Ré  
*Ask Me Anything: A simple strategy for prompting language models*  
 International Conference on Learning Representations (ICLR) 2023  
**Spotlight Award (Top 5% of 5K submissions)**  
[Paper Link](#) / [GitHub Link](#)
- [20] Simran Arora, Sen Wu, Enci Liu, Christopher Ré  
*Metadata shaping: A simple approach for knowledge-enhanced language models*  
 Findings of the Association for Computational Linguistics (ACL) 2022  
[Paper Link](#) / [GitHub Link](#)
- [21] Avanika Narayan, Laurel Orr, Ines Chami, Simran Arora, Christopher Ré  
*Can Foundation Models Wrangle Your Data?*  
 Proceedings of the VLDB Endowment (PVLDB) 2022  
[Paper Link](#) / [GitHub Link](#)
- [22] Laurel Orr, Megan Leszczynski, Simran Arora, Sen Wu, Neel Guha, Xiao Ling, Christopher Ré  
*Bootleg: Chasing the Tail with Self-Supervised Named Entity Disambiguation*  
 Conference on Innovative Data Systems Research (CIDR) 2021  
[Paper Link](#) / [GitHub Link](#)
- [23] Simran Arora, Avner May, Jian Zhang, Christopher Ré  
*Contextual Embeddings: When Are They Worth It?*  
 Proceedings of the Association for Computational Linguistics (ACL) 2020  
[Paper Link](#) / [GitHub Link](#)
- [24] Qizhen Zhang, Akash Acharya, Hongzhi Chen, Simran Arora, Ang Chen, Vincent Liu, Boon T. Loo  
*Optimizing Declarative Graph Queries at Large Scale*  
 Proceedings of the International Conference on Management of Data (SIGMOD) 2019  
[Paper Link](#)
- [25] Edward Steager, Denise Wong, Jeremy Wang, Simran Arora, Vijay Kumar  
*Control of multiple microrobots with multiscale magnetic field superposition*  
 Intl. Conference on Manipulation, Automation and Robotics at Small Scales (MARSS) 2017  
**Best Paper Runner Up (Top 3 papers)**  
[Paper Link](#)
- [26] B. P. Mason, M. Whittaker, J. Hemmer, Simran Arora, A. Harper, S. Alnemrat, A. McEachen, S. Helmy, J. Read de Alaniz, J. P. Hooper  
*A temperature-mapping molecular sensor for polyurethane-based elastomers*  
 Applied Physics Letters (APL) 2016  
[Paper Link](#)

## Workshop

- [1] Willy Chan, Nathan Paek, Simon Guo, Simran Arora, Daniel Y Fu  
*ParallelKernelBench: Can LLMs Write Fast Multi-GPU Kernels?*

ICML DL4C 2026

[Paper Link](#) / [GitHub Link](#)

- [2] Arya Tschand, Kesavan Ramakrishnan, Alexander Ingare, Simon Guo, Jeffrey Jian Ma, Zishen Wan, Simran Arora, Azalia Mirhoseini, Vijay Janapa Reddi

*Hawkeye: Hardware-Aware GPU Kernel Optimization with Minimal Supervision*

ICML DL4C 2026 **Oral Award (Top 5 of 80+ accepted papers)**

[Paper Link](#)

- [3] Simran Arora, Aman Timalsina, Aaryan Singhal, Benjamin Spector, Sabri Eyuboglu, Xinyi Zhao, Ashish Rao, Atri Rudra, and Christopher Ré

*Just read twice: closing the recall gap for recurrent language models*

ICML ES-FoMo 2025

[Paper Link](#) / [GitHub Link](#)

- [4] Sabri Eyuboglu, Dylan Zinsley, Jon Saad-Falcon, Simran Arora, Atri Rudra, James Zou, Chris Ré

*Towards smaller language models via layer looping*

ICML ES-FoMo 2024

[Paper Link](#)

- [5] Simran Arora and Christopher Ré

*Can Foundation Models Help Us Achieve Perfect Secrecy?*

AAAI PPAI Workshop 2023

[Paper Link](#) / [GitHub Link](#)

## Invited Talks

|      |                                                                                         |
|------|-----------------------------------------------------------------------------------------|
| 2025 | Scale ML x GPUMode Lecture ( <a href="#">Link</a> ; Virtual)                            |
| 2025 | Stanford CS229S Guest Lecture (Stanford, CA)                                            |
| 2025 | MLSys Conference YPS Keynote Speaker (Santa Clara, CA)                                  |
| 2025 | University of Toronto (Toronto, Canada)                                                 |
| 2025 | Massachusetts Institute of Technology (Cambridge, MA)                                   |
| 2025 | New York University (NYC, NY)                                                           |
| 2025 | Princeton University (Princeton, NJ)                                                    |
| 2025 | University of Pennsylvania (Philadelphia, PA)                                           |
| 2025 | University of Washington (Seattle, WA)                                                  |
| 2025 | UCLA CS & ECE (Los Angeles, CA)                                                         |
| 2025 | University at Buffalo (Buffalo, NY)                                                     |
| 2025 | Columbia CS & EE (NYC, NY)                                                              |
| 2025 | Cornell University (Ithaca, NY)                                                         |
| 2025 | Cornell Tech (NYC, NY)                                                                  |
| 2025 | California Institute of Technology (Pasadena, California)                               |
| 2025 | Rice University (Houston, Texas)                                                        |
| 2025 | Northwestern University (Evanston, Illinois)                                            |
| 2025 | University of Michigan (Ann Arbor, Michigan)                                            |
| 2025 | NVIDIA GTC @ GPU Mode (San Jose, CA)                                                    |
| 2025 | NVIDIA (Virtual)                                                                        |
| 2024 | NeurIPS Hacker-Cup AI Workshop Keynote Speaker ( <a href="#">Link</a> ; Vancouver, BC)  |
| 2024 | Simons Institute: Are Transformers the end game? ( <a href="#">Link</a> ; Berkeley, CA) |

## Invited Talks Cont'd

|      |                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 2024 | Simons Institute: Transformers as a Computational Model ( <a href="#">Link</a> ; Berkeley, CA)                        |
| 2024 | Stanford NLP Group (Stanford, CA)                                                                                     |
| 2024 | UC Berkeley NLP Group (Berkeley, CA)                                                                                  |
| 2024 | CCAIM Summer School ( <a href="#">Link</a> ; Virtual)                                                                 |
| 2024 | Liquid AI (Vienna, Austria)                                                                                           |
| 2024 | Princeton University PLI Group ( <a href="#">Link</a> ; Princeton, NJ)                                                |
| 2024 | Cornell Tech (New York, NY)                                                                                           |
| 2024 | Microsoft AI Research (Virtual)                                                                                       |
| 2024 | 56th Annual ACM Symposium on Theory of Computing Workshop Keynote Speaker ( <a href="#">Link</a> ; Vancouver, Canada) |
| 2023 | NeurIPS 3rd Table Representation Learning Workshop Keynote Speaker ( <a href="#">Link</a> ; New Orleans, LA)          |
| 2023 | Snorkel Foundation Model Summit (Virtual)                                                                             |
| 2023 | Apple Machine Learning Research (Cupertino, CA)                                                                       |
| 2022 | IBM AI Research (Virtual)                                                                                             |
| 2022 | MIT Computational Social Science (Virtual)                                                                            |
| 2021 | Facebook AI Research (Virtual)                                                                                        |
| 2021 | Spotlight at Stanford HAI Data-Centric AI Workshop (Virtual)                                                          |
| 2020 | Stanford DAWN Retreat (Virtual)                                                                                       |

## Teaching

|             |                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summer 2025 | <b>Course Assistant</b> , <i>CS 229: Machine Learning</i><br>Stanford University                                                                                            |
| Fall 2023   | <b>Course Co-Creator and Co-Instructor</b> , <i>CS 229S: Systems for Machine Learning</i><br>Stanford University<br>3-Unit undergrad/graduate course, Taught 110+ students. |
| Fall 2023   | <b>Instructor</b> <i>CS 528: Machine Learning Systems Seminar</i><br>Stanford University                                                                                    |
| Spring 2019 | <b>Course Co-Creator</b> , <i>MCIT 595: Computer Systems</i><br>University of Pennsylvania                                                                                  |
| Fall 2018   | <b>Course Assistant</b> , <i>CIS 380: Operating Systems</i><br>University of Pennsylvania                                                                                   |
| Spring 2018 | <b>Course Assistant</b> , <i>CIS 160: Discrete Mathematics</i><br>University of Pennsylvania                                                                                |
| Fall 2017   | <b>Course Assistant</b> , <i>CIS 160: Discrete Mathematics</i><br>University of Pennsylvania                                                                                |

## Service

|           |                                                                                             |
|-----------|---------------------------------------------------------------------------------------------|
| Ongoing   | ICML (Top Reviewer Award), NeurIPS, ACL, PPAI-AAAI, NeurIPS TRL, ICLR ME-FoMo, ICML ES-FoMo |
| 2024-2025 | Looma AI Volunteer (and 2013-2015)                                                          |
| 2025      | ICML ES-FoMo Workshop Organizer ( <a href="#">Link</a> )                                    |
| 2025      | NeurIPS Efficient Reasoning Workshop Organizer ( <a href="#">Link</a> )                     |
| 2022-2023 | East Palo Alto Academy Foundation Volunteer                                                 |